FINAL PROJECT SUBMISSION

MIRQAB

AI-Powered Security Operations Platform for SMBs — a Security Operations Center (SOC) platform that turns security events into investigated, explainable incidents with a timed, human-controlled response.

TEAM

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PROJECT

MIRQAB — AI-Powered Security Operations Platform for SMBs
27 September 2026

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How to read the status labels

VERIFIED	Tested and confirmed: by the automated test suite (38 suites, 421 tests, all passing on 27 Sep 2026) and/or by a live test recorded in the team's engineering reports.
VERIFIED IN LAB	Confirmed end-to-end in a controlled, containerized lab. Not tested on physical employee devices.
LIMITED	Implemented and working, with a clearly stated limitation.
NOT VERIFIED	Implemented in code but not tested live.
NOT IMPLEMENTED	Not available in the application.
NOT DEPLOYED	No production deployment exists.
BLOCKED	Cannot be completed because of an external dependency.
PLANNED	A future roadmap item. Not built.
DEMO / SIMULATION	Uses synthetic or public demo data. Not a real environment.

1. Project at a glance

Project	MIRQAB — AI-Powered Security Operations Platform for SMBs
Team	Bdoor Fahed Alosaimi · Nouf Saleh Altalhi · Alaa Rafa Al Saad
Type	A web-based Security Operations Center (SOC) platform, plus a desktop endpoint agent
Applications	SOC Dashboard (web) · Employee Desktop (Electron) · shared backend API
Core stack	TypeScript · NestJS · React · PostgreSQL · Prisma · Redis · Socket.IO · nginx · Docker · Wazuh
State on 27 Sep 2026	Working prototype running locally in Docker; 421 automated backend tests passing; NOT DEPLOYED to production

2 The problem

Small and medium-sized businesses (SMBs) face the same attacks as large enterprises — phishing, credential brute-force, ransomware, command-and-control beacons — but rarely have a staffed Security Operations Center.

- **Too many raw alerts, too little context.** Tools such as Wazuh or Suricata produce low-level events. Turning them into an understandable incident — what happened, on which device, how serious, what to do — takes expertise an SMB often lacks.
- **Slow response.** Without a clear owner and a deadline, a high-severity alert can sit unacknowledged while an attacker moves on.
- **Unsafe automation.** Fully automatic "block everything" responses can disrupt the business; purely manual responses can be too slow. SMBs need automation that is bounded, explainable and auditable.
- **Data sensitivity.** Sending internal security logs to an external AI service is often unacceptable for regulated or privacy-conscious organizations.

3. Target users

User	Role in MIRQAB	What they need
SMB owner / IT manager	`ADMIN`	Organization-wide visibility, user management, policies, reports.
Security analyst (in-house or outsourced)	`SECURITY_ANALYST`	Investigate incidents, approve responses, track tickets.
IT technician	`IT_TECHNICIAN`	Act on incidents in their own department only.
Employee	`EMPLOYEE`	Device health and security guidance through the Employee Desktop app (no SOC dashboard access).

4. Why MIRQAB exists

MIRQAB (Arabic: مرقاب, "watchtower") was built to give an SMB the core workflow of a SOC without a SOC-sized team: every event becomes an explainable incident, every high-severity incident gets a clock, and every response — human or automated — is authorized, confirmed and recorded. It deliberately favors deterministic, auditable logic and keeps AI inside a strict, optional boundary.

5. Proposed solution

A single platform that:

1. **Ingests** security events from Wazuh, Suricata and Sysmon forwarders (plus public datasets and a Simulation Mode for demonstration).
2. **Detects and enriches** them with deterministic rules: severity, MITRE ATT&CK technique, indicators of compromise/attack (IOC/IOA), persistence and command-and-control patterns.
3. **Correlates** related events into one case with a Case ID, evidence and timeline.
4. **Guides the investigation** with similar past incidents, risk statistics and Suggested Solutions.

5. **Enforces a 5-minute response window** for HIGH/CRITICAL incidents, with a single, once-only automated-containment decision if nobody responds.
6. **Contains** a compromised endpoint through Wazuh Active Response, trusting only an agent-confirmed result.
7. **Reports** through per-case PDF reports and a scheduled weekly SOC report.

6 Main capabilities

Detection & enrichment

Rule-based classification, MITRE mapping (12 curated techniques), IOC/IOA extraction, persistence and C2 detection. VERIFIED

Correlation & cases

Events merge into open cases (72-hour window, similarity score ≥ 65), each with a Case ID such as `INC-20260919-0001`. VERIFIED

Investigation workspace

Two-window incident view: *Incident Response* (actions) and *Incident Investigation* (what happened, evidence, MITRE, indicators, analysis). VERIFIED

Suggested Solutions

Technique-specific fixes plus actions that worked on similar past incidents; high-impact actions require approval. VERIFIED

5-minute response window

Countdown, acknowledgement and once-only containment decision for HIGH/CRITICAL incidents. VERIFIED (automated success live: LIMITED)

Device isolation

Wazuh Active Response with agent-confirmed outcome. VERIFIED IN LAB

Live network map / Digital Twin

Topology of departments, devices and active attacks; "The Citadel" live view and APT29 attack-path view. VERIFIED

Reports

Case report PDF per incident; weekly SOC report PDF (scheduled + on demand). VERIFIED

Security & governance

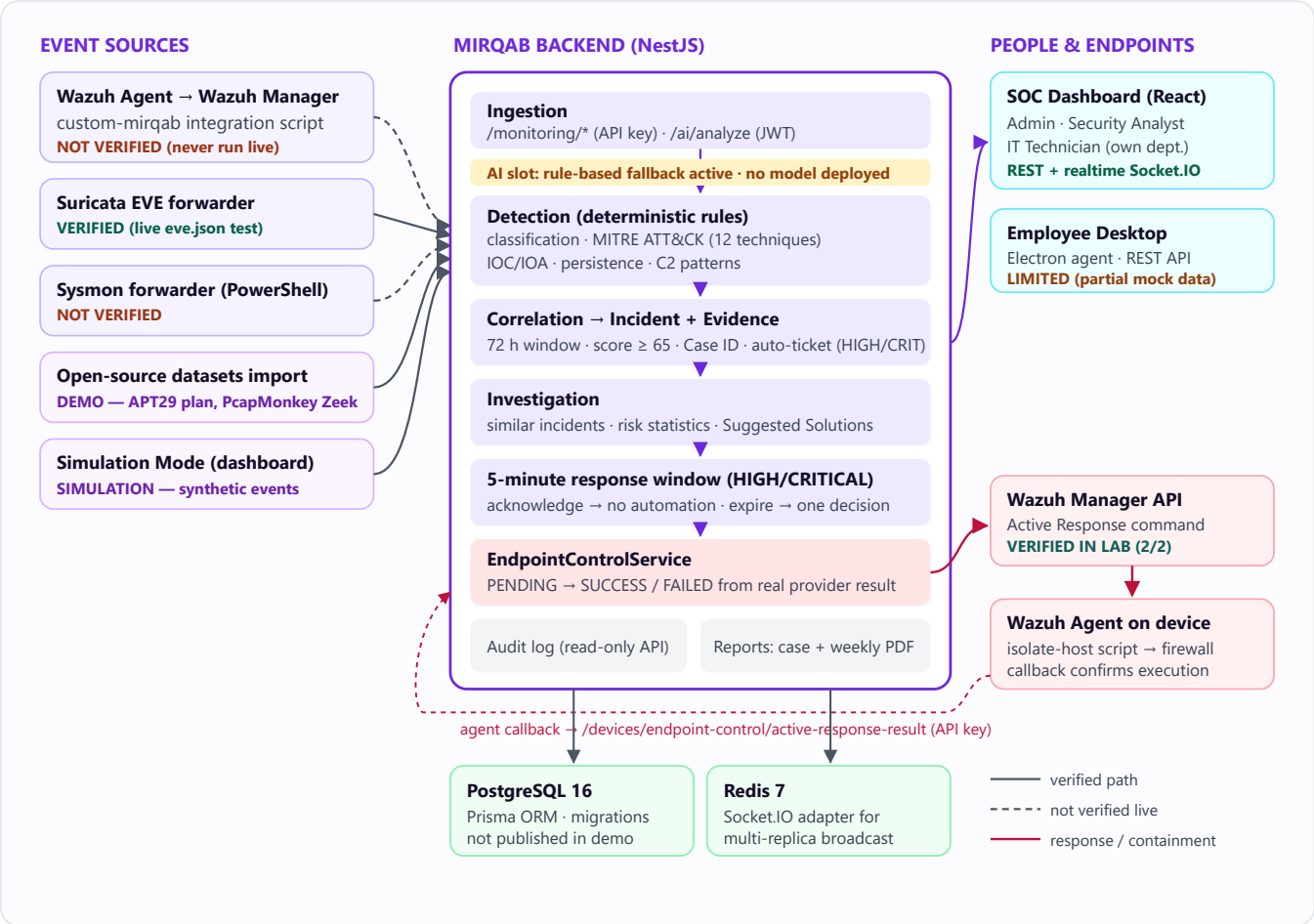
JWT auth, 4-role RBAC, department and organization scoping, read-only audit log, rate limiting. VERIFIED

Bilingual interface

Arabic (right-to-left) and English, dark and light themes. LIMITED (some labels untranslated)

7 Architecture overview

MIRQAB is a TypeScript monorepo: a NestJS backend, a React SOC Dashboard, an Electron Employee Desktop app, and Docker Compose infrastructure (PostgreSQL, Redis, nginx). The diagram shows the **actual** data flow; dashed links are implemented but not verified live.



Note: MIRQAB does not read from the Wazuh Indexer. Wazuh alerts reach MIRQAB through a Manager-side integration script (not verified live) or through the verified ingestion endpoint. See the Technical Architecture document for details.

8 SOC workflow

1. **Event arrives** at `POST /monitoring/wazuh|suricata|sysmon` (shared API key) or `POST /ai/analyze` (authenticated).
2. **Classification** assigns severity and a MITRE technique; enrichment extracts indicators, persistence and C2 findings.
3. **Correlation** either adds the event as evidence to an existing open case or creates a new incident with a Case ID.
4. **HIGH/CRITICAL incidents** automatically get a ticket, a "response required" notification and a 5-minute response window.
5. **The analyst investigates** in the two-window incident view: evidence, MITRE context, indicators, similar incidents, Suggested Solutions.
6. **The analyst responds**: acknowledge, escalate, collect evidence, isolate the endpoint (with confirmation), log actions, update status.
7. **Everything is recorded** in the audit log and incident timeline, and summarized in the case report and weekly report.

9. The 5-minute response workflow

Step	What MIRQAB does
T0 — detection	A HIGH or CRITICAL incident is created. The deadline is T0 + 5 minutes. MEDIUM/LOW incidents never get automated containment — they always wait for an analyst.
Notification	A "response required" notification goes to the SOC users who can see the incident (Admin, Security Analyst, and the IT Technician of that department).
Acknowledgement	"Acknowledge & Investigate" (or any status change) records the response. Acknowledging is idempotent; once acknowledged before the deadline, automated containment never happens for that incident.
Deadline expires	A background sweep takes **one, once-only decision** inside a per-incident database lock: if the incident is simulated, the "5-Minute Response Window" policy is disabled, or no endpoint-control provider is configured, containment is recorded as *not attempted*; otherwise a PENDING `ISOLATE_DEVICE` action is saved first, the provider is called, and the same record becomes SUCCESS or FAILED from the real result.
Late response	An acknowledgement after the deadline is recorded as a late response; it never silently undoes an isolation.

Status

The workflow logic is VERIFIED by unit tests and shown in the UI. A **successful automated isolation against a live agent has not been demonstrated**: in the development environment automated attempts were recorded as FAILED because no reachable endpoint was behind the device. LIMITED

10 Wazuh integration

Direction	Mechanism	Status
Wazuh → MIRQAB (alerts)	Ingestion endpoint `POST /monitoring/wazuh`, normalizing Wazuh alerts into the common detection path.	VERIFIED with Wazuh-format payloads
Wazuh Manager forwarding	Custom integration script (`custom-mirqab`) invoked by the Manager for each alert.	NOT VERIFIED
Log enrichment	Optional Wazuh *logtest* enrichment through the Manager API.	LIMITED — disabled unless Wazuh API credentials are set
MIRQAB → Wazuh (containment)	Active Response command through the Manager API; agent script applies firewall isolation and calls back to confirm.	VERIFIED IN LAB
Wazuh version	Targets 4.14.7; the upstream image was a release candidate when checked.	BLOCKED

11. Incident management

Each incident has a Case ID, severity, status (Open → Investigating → Contained → Resolved → Closed), device, department, evidence items with their source ("Live Agent", "Open-Source Dataset", "Simulation (Synthetic)", "Manual / Test"), a timeline, tickets, response actions with outcomes, and a resolution record (root cause, final resolution, lessons learned). HIGH/CRITICAL incidents automatically open a ticket.

12. MITRE ATT&CK mapping

A curated reference of **12 techniques** — T1566 Phishing, T1110 Brute Force, T1204 User Execution, T1046 Network Service Discovery, T1068 Exploitation for Privilege Escalation, T1486 Data Encrypted for Impact, T1071 Application Layer Protocol, T1053 Scheduled Task/Job, T1078 Valid Accounts, T1059 Command and Scripting Interpreter, T1105 Ingress Tool Transfer, T1567 Exfiltration Over Web Service — is matched deterministically against event text. Unmatched events are honestly labelled `UNCLASSIFIED`. If an AI provider were enabled, its output would be validated against this reference to reject invented technique IDs.

13. IOC and evidence handling

- **Indicators of compromise (IOC):** IP addresses, file hashes, domains, URLs and email addresses, each with a confidence value.
- **Indicators of attack (IOA):** behaviour patterns — credential dumping, lateral movement, process injection, obfuscated execution, defense evasion.
- **Persistence and C2 findings:** scheduled tasks, services, run keys, beacons and similar patterns, mapped to MITRE.
- **Evidence:** every contributing event is stored with its raw excerpt, source label and correlation score; nothing is fabricated — no match means no indicator.

14 Device isolation

- Only authorized roles on an incident they can access can request isolation; the target device is resolved **on the server** from the incident, never taken from the request.
- The analyst confirms in a dialog (only this endpoint is affected; recorded against the case).
- The action is saved as PENDING, then MIRQAB calls Wazuh Active Response. The outcome becomes SUCCESS only when the agent's own callback confirms execution — a "dispatched" reply from the Manager is never trusted alone. Failures are recorded as FAILED with the reason.
- A device shows as "Isolated" only when a genuine SUCCESS isolation action exists.
- AI components can never trigger isolation: no AI/agent code path is wired to the endpoint-control service.

Status

VERIFIED IN LAB — 2/2 end-to-end isolations on a containerized Linux safe-test agent (11–15 s). Not tested on physical devices; Windows script NOT VERIFIED ; releasing isolation NOT IMPLEMENTED in the application.

15. Notifications

In-app notifications (bell icon and Notifications page) are delivered in real time over Socket.IO: "response required" for new HIGH/CRITICAL incidents and "containment result" when the window expires. Recipients are exactly the SOC users allowed to see the incident. LIMITED : there is no per-employee notification — the backend has no employee-ownership model for incidents.

16. Digital Twin / live network map

The Overview page and the *Live Network Map* render the organization as a topology — internet, firewall, core switch, application and database servers, departments and their devices — with active attacks highlighted; devices open their incidents directly. *The Citadel — Live View* is a wallboard-style view, including an **Attack Path** tab that draws the public APT29 emulation plan (DEMO data). The topology is built from real device, department and incident records; it is a representation of MIRQAB's own data, not a live network scanner.

17. Reports

- **Case report (PDF):** per incident, 11 sections — incident summary, evidence, timeline, IOC, IOA, persistence findings, command-and-control, AI analysis (agent decisions and risk statistics), response actions, recommendations and audit trail. Access-controlled like the incident. VERIFIED
- **Weekly SOC report (PDF):** generated automatically every Monday 05:00 UTC for the just-completed Sunday–Saturday week (per organization), or on demand; downloadable from the Reports page. VERIFIED

18 AI architecture

MIRQAB's "AI" layer is deliberately conservative:

- **Current behaviour:** 100% deterministic, rule-based analysis. No language model is running. Features labelled as "agents" in the interface (Incident Agent, IOC/IOA Agent, MITRE Risk Agent, Recommendation Agent) are named, least-privilege wrappers around deterministic services.
- **Provider slot:** one AI provider can be configured at a time (never chained). With none configured, every AI-eligible task uses its rule-based fallback.
- **Safety boundary:** prompt-injection delimiting, strict output validation (severity and MITRE IDs checked against the reference), circuit breaker, timeouts and a concurrency cap. AI output can inform an analyst; it can never execute a response.
- **External AI:** an Anthropic Claude integration exists for development comparison only. It requires both an API key **and** an explicit `ALLOW_EXTERNAL_AI_DEV_ONLY=true` flag; it is disabled by default and in the evaluator demo. NOT USED AT RUNTIME

19. Self-hosted / private AI direction

The intended production direction is a **self-hosted model inside the customer's own network** (`SELF_HOSTED_AI_URL`), so security logs never leave the organization. The provider interface, data minimization, validation and fail-closed defaults are implemented and tested. NOT VERIFIED / NOT DEPLOYED: no model has been deployed — the development laptop lacks the hardware to run one safely alongside the stack.

20. Security model

- **Authentication:** JWT access tokens (15 min) pinned to HS256, rotating refresh tokens (7 days) with reuse detection, bcrypt (12 rounds), generic login errors, rate-limited auth endpoints (5 requests/min).
- **Authorization:** 4 roles; IT Technicians are limited to their department; every record is scoped to the user's organization; object-level guards on incidents, tickets and devices.
- **API hardening:** strict input validation (unknown fields rejected), UUID validation, null-byte rejection, length limits, Helmet security headers, API documentation disabled in production.
- **Realtime:** Socket.IO handshake authenticated with the same JWT; dashboard roles only; department-scoped rooms; no client-to-server messages; per-IP connection limit; disconnect at token expiry.
- **Audit:** server-derived actor on every action, denied and failed-login attempts logged, audit API is read-only.
- **Infrastructure:** nginx TLS termination with security headers; backend container runs as a non-root user; secrets only in untracked `.env` files.

21 Verified security work

Between 6 and 8 September 2026 the team ran an 18-phase security program, each phase ending in a written report. Summary:

Phase	Area	Outcome
1	RBAC / authorization / IDOR-BOLA	VERIFIED — 18/18 live tests
2 / 2A	Device isolation / Active Response reliability	VERIFIED IN LAB — 2 script bugs fixed, 2/2 end-to-end
3	Secrets and credentials	VERIFIED — secrets rotated, git exposure fixed
4	Wazuh version stability	BLOCKED — no GA image available
5	API security (OWASP API)	VERIFIED — 3 findings fixed
6 / 7	Audit logging, incident and alert integrity	VERIFIED
8	Self-hosted / private AI	NOT VERIFIED — no local model hardware
9	AI failure, fallback and reliability	VERIFIED
10	Dependency, container and supply chain	VERIFIED
11	Authentication, session, rate limiting	VERIFIED
12	Realtime / Socket.IO	VERIFIED
13	Frontend security	VERIFIED
14	Live map / Digital Twin	VERIFIED
15	Production failure modes	VERIFIED WITH LIMITATIONS — some scenarios static only
16	Performance and scale	VERIFIED WITH LIMITATIONS — no load test
17 / 18	Final regression and acceptance	VERIFIED WITH DOCUMENTED LIMITATIONS

Re-checked on 27 Sep 2026: the backend test suite now has 38 suites / 421 tests (293 at Phase 18), all passing; all three applications type-check cleanly. Features added after 8 Sep have unit-test coverage but no dedicated security review.

22 Current implementation status

Capability	Status	Evidence and notes
Overall deployment	NOT DEPLOYED	MIRQAB runs locally in Docker on the team's development machine. There is no production deployment, public domain or CA-issued TLS certificate (a self-signed development certificate is used).
Backend API (NestJS)	VERIFIED	38 test suites / 421 tests passing and a clean TypeScript check on 27 Sep 2026. Running locally in Docker.
SOC Dashboard (React web app)	VERIFIED	Clean TypeScript check. Rebuilt from the current source and served at `https://localhost/` on 27 Sep 2026.
Employee Desktop agent (Electron)	LIMITED	Clean TypeScript check. Reads real device telemetry. Some screens (AI scan, sample incident history) show clearly-marked mock data. Not packaged or distributed.
Authentication and sessions	VERIFIED	JWT pinned to HS256, refresh-token rotation with reuse detection, bcrypt password hashing, rate-limited login. Unit tests plus live tests (security Phase 11).
Role-based access control and department scoping	VERIFIED	4 roles. 18 of 18 live cross-department / IDOR tests passed (security Phase 1), plus unit tests.
Organization (tenant) isolation	VERIFIED LIMITED	Enforced in the access guards and in every scoped query, and covered by unit tests. No live test with several organizations is recorded.
Wazuh alert ingestion endpoint	VERIFIED	`POST /monitoring/wazuh` tested live with Wazuh-format alert payloads. Each one created a real incident.
Wazuh Manager integration script	NOT VERIFIED	The `custom-mirqab` script that forwards alerts from a Wazuh Manager was reviewed but never run against a real manager.
Suricata EVE forwarder	VERIFIED	Run live against a growing `eve.json` file. It created a real incident.
Sysmon forwarder	NOT VERIFIED	Sysmon was not installed on the development machine.
Device isolation via Wazuh Active Response (Linux script)	VERIFIED IN LAB	2 of 2 end-to-end isolations through the MIRQAB application against a containerized safe-test Wazuh agent. A real firewall rule was applied and confirmed by an agent callback, measured at 11–15 s (security Phase 2A). Not tested on physical devices.
Windows isolation script	NOT VERIFIED	`isolate-host.ps1` exists but has not been run against a real Windows agent.
Releasing an isolated device (un-isolate)	NOT IMPLEMENTED	The application has no "release isolation" action. A companion script exists but was never deployed or tested.

Capability	Status	Evidence and notes
Wazuh general-availability version 4.14.7	BLOCKED	When checked (security Phase 4), the upstream 4.14.7 image tag still served release-candidate builds. The lab used that build.
5-minute response window	VERIFIED	Countdown, acknowledgement and once-only decision are covered by unit tests. The UI is shown in QA screenshots from 22–26 Sep 2026.
Automated containment when the window expires	LIMITED	The decision logic is unit-tested. In the development environment, automated attempts were correctly recorded as **FAILED** ("fetch failed") because no reachable endpoint was behind the device. A successful automated isolation against a live agent is not recorded.
Simulation safety (no real isolation for simulated incidents)	VERIFIED	Incidents whose source is `SIMULATION` are excluded from automated containment (unit-tested). The evaluator demo also hard-codes every endpoint-control setting to empty.
MITRE ATT&CK mapping	VERIFIED LIMITED	Deterministic, rule-based mapping to 12 curated techniques, unit-tested. Events that match no rule stay `UNCLASSIFIED` (11 of 27 demo incidents).
IOC / IOA extraction	VERIFIED	Deterministic extraction of IPs, hashes, domains, URLs and emails, plus 5 behaviour patterns. Nothing is invented: no match means no indicator.
Incident correlation	VERIFIED	Events join an open case within a 72-hour window when their similarity score is 65 or higher. Unit-tested.
Incident memory and similar-incident search	VERIFIED	Native vector similarity (equivalent to a FAISS flat inner-product index), department-scoped. Unit-tested.
Risk prediction	LIMITED	Statistics calculated from the department's own past incidents. It is not a machine-learning forecast, and it returns "insufficient data" when there is no history.
Suggested Solutions	VERIFIED	Fixes from a per-technique catalog plus actions that worked on similar past incidents. High-impact actions need analyst approval. Unit-tested.
External AI (Anthropic Claude)	NOT USED AT RUNTIME	Disabled by default. It is a development-only comparison option that needs both an API key and an explicit dev-only flag. It is off in the evaluator demo.
Self-hosted / private AI provider	NOT VERIFIED NOT DEPLOYED	The provider interface is implemented and tested, but no local model has been deployed (no suitable hardware). All AI-labelled analysis currently uses deterministic rules.
Notifications for SOC staff	VERIFIED	"Response required" and "containment result" notifications are scoped like the incidents they describe. Unit-tested.

Capability	Status	Evidence and notes
Notifications to individual employees	LIMITED	The backend has no per-employee ownership model. Notifications go to SOC roles (Admin, Security Analyst, IT Technician).
Live network map / Digital Twin	VERIFIED	Topology is built from real device, department and incident records, and its security was reviewed (security Phase 14). The APT29 attack-path view uses static public dataset data DEMO.
Case report (PDF)	VERIFIED	Generated per incident. Unit-tested, including access control.
Weekly SOC report (PDF)	VERIFIED	Scheduled every Monday at 05:00 UTC, plus manual generation. Unit-tested.
Arabic / English interface	LIMITED	Full language switch with right-to-left layout, but some incident-panel labels still appear in English in Arabic mode.
Dark / light theme	VERIFIED	Both themes are shown in QA screenshots.
Evaluator demo stack (`infra/demo`)	VERIFIED LIMITED	On 27 Sep 2026 the full demo stack was run end-to-end from the packaged source zip, using the documented one command, on the development machine. All 5 services were healthy, 22 + 6 dataset events were imported, and all 48 checks passed (API, RBAC, realtime, reports, browser walkthrough, response window). **It has not been run on a separate clean machine.**
Load / performance testing	NOT VERIFIED	No load test was performed. Query-count improvements were verified with unit tests only (security Phase 16).

23 Known limitations

1. **Not deployed to production.** MIRQAB runs locally in Docker on the development machine. It has no public domain, and TLS uses a self-signed development certificate.
2. **No runtime AI model.** Classification, MITRE mapping, risk estimates and recommendations are deterministic and rule-based. The self-hosted AI provider is implemented but **not verified**: no model has been deployed. External AI (Claude) is disabled by default and is never used in the demo.
3. **Device isolation was verified only in a lab.** 2 of 2 successful end-to-end isolations used a containerized Linux safe-test agent. There are no tests on physical employee devices, the Windows script is not verified, and releasing an isolated device is not implemented in the application.
4. **Automated containment has not been shown to succeed live.** When the 5-minute window expires with no response, MIRQAB records an isolation request. In the development environment those attempts were recorded as **FAILED**, because no reachable endpoint existed behind the device.
5. **Wazuh version is BLOCKED.** The upstream 4.14.7 image was still a release-candidate build when checked. The lab ran that build.
6. **Wazuh Manager → MIRQAB forwarding script not verified.** The ingestion endpoint was verified with Wazuh-format payloads, but the Manager-side script was never run. MIRQAB does **not** read from the Wazuh Indexer.
7. **Limited detection coverage.** 12 curated MITRE ATT&CK techniques. Events that match no rule stay `UNCLASSIFIED`. For example, the dataset notes report all 6 PcapMonkey network events landing as UNCLASSIFIED / LOW.
8. **Notifications target SOC staff, not individual employees.** The Employee Desktop app partly uses clearly-marked mock data.
9. **Partial Arabic translation** in some incident-panel labels.
10. **No load testing**, and no live multi-organization test.
11. **No automated backup or disaster-recovery tooling.** Backups are manual (`pg_dump`).
12. **Demo setup not run on a clean machine.** The full evaluator demo was run end-to-end from the packaged source (48 of 48 checks passed), but only on the development machine, where Docker and base images were already present.
13. **Security review dates.** The 18-phase security review was completed on 6–8 Sep 2026. Features added later (the reworked 5-minute response window, Suggested Solutions, the evaluator demo) are covered by the current unit tests but have not had a dedicated security review.

24. Future roadmap

Proposed next steps, derived from the documented limitations and deferred items. All are **PLANNED** — none is built.

1. Deploy and evaluate a self-hosted language model on suitable hardware inside the customer network.
2. Validate isolation on physical Windows and Linux endpoints; add a "release isolation" action.
3. Move to a general-availability Wazuh release once published; run the Manager integration script end-to-end.
4. Production deployment with a real domain and CA-issued TLS certificate; automated backups and a disaster-recovery procedure.
5. Broaden detection coverage beyond the 12 curated techniques; load and performance testing.
6. Complete Arabic translation; employee-level notifications through the Employee Desktop app.

7. A dedicated security review of the features added after 8 September 2026.